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RESEARCH ARTICLE

Early Warning of Student Performance With Integration of Subjective and Objective Elements

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ABSTRACT Early warning of student performance is using data analytics to predict future trends in student performance as a way to intervene early in situations of academic risk. The popularity of machine learning has improved prediction accuracy. However, most current models only consider subjective student factors without examining the external environment and objective elements. Meanwhile, the global pandemic of COVID-19 has brought serious disruptions to teaching and learning in universities, and existing models cannot cope with this challenge. In this study, we propose a neural network model that integrates various internal and external factors and incorporates data-level sample synthesis and multi-classification cost-sensitive learning methods to achieve early warning of student performance in universities and improve teaching quality and management. Experimental results show that the model can be applied to teaching scenarios with a mixture of online and offline teaching, has higher accuracy than previous prediction mechanisms that only consider some student's academic characteristics, and outperforms traditional machine learning methods.

INDEX TERMS Education data mining, deep learning, performance alert, predictive model, online education.

I. INTRODUCTION

The continuous development and improvement of information technology construction in higher education institutions and related businesses has collected a large amount of data on students' academic performance, which has played an important role in promoting educational innovation and development [1]. In addition, the concepts of smart campuses and networks have been proposed in recent years. In addition to the data relevant to the business side, many behaviors have been collected. IoT logs and accumulated big data also provide a good basis for applying data-driven technology in student academic alerts. Many scholars have focused on the tremendous social value of big data in education, and have researched an early academic warning. According to the existing research results, the use of various data analytics for educational performance prediction and analysis will bring the following values:

- Self-assessment of students' academic ability to adjust learning strategies and manage learning progress in advance.
- Teachers and other educators use data analysis to understand teaching and learning and receive positive intelligence feedback about students and subjects with academic problems. Teaching and learning can be improved through early intervention.
- Higher education administrators use teaching data analysis and insight to predict teaching quality and student academic level trends in advance and support relevant administrators in the subsequent development.

In China, the enrollment scale of higher education has been steadily expanding in the past decade, and the trend of enrollment growth is obvious. More students have the opportunity to enter universities for higher education [2]. However,

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on the other hand, this has raised concerns about the quality of university teaching. Problems with faculty allocation and distribution of teaching resources have led to the poor quality of some students completing their courses. With the huge number, the number of students with academic problems cannot be ignored. To regulate and improve the quality of teaching and graduates in universities, the government has introduced several policies and regulations in recent years. As a result, researchers have also proposed mechanisms to track and warn students' performance to intervene earlier to ensure the quality of teaching and graduation.

In most of the existing studies, researchers have always taken more into account students' grades and other personal background factors (e.g., family environment, income) [3], [4], or have only studied the grades of an individual course in a given academic year, with little analysis of the difficulty of the course itself and the teaching of the course. In colleges and universities, the same course varies in importance, credit and instructional status, and the content of instruction and assessment in different colleges.

Therefore, unlike existing studies, this study considers not only the characteristics of the distribution of students' grades and their daily activities in school, but also the impact of different course difficulties, instructors, and credit differences on final grades over a longer time. Obviously, these factors are not within the student's control, but their impact on student achievement cannot be ignored.

In addition, due to the impact of the COVID-19 global pandemic, the traditional offline teaching model has been disrupted and students have less time for mutual contact with the instructor, thus increasing the lack of discussion and effective interaction with the instructor when faced with learning difficulties. Which some researchers are more concerned about the impact on teaching quality [5]. Psychological changes in students also affect student learning outcomes and outputs [6]. At the same time, unfortunately, the data used in existing prediction models will become inaccurate in this scenario, leading to prediction failure, while the performance warning mechanism specifically for MOOC or online courses cannot be applied to the offline environment.

Therefore, this study attempts to consider the impact of the COVID-19 on teaching and learning in universities as an external environmental characteristic and to consider the impact on prediction results in a comprehensive manner. In summary, the main contributions of this study are as follows:

- A new deep learning model SAL&EC(Subjective Ability Level & External Characteristics) neural network is proposed to provide early warning for students with academic problems in colleges and universities, based on existing research and taking into account a variety of factors.
- A new perspective of optimizing the accuracy of the model is proposed, i.e., considering the performance of historical courses and other students, breaking the previous status quo of analyzing only the individual student's situation.
- The current situation of teaching and learning in universities under the influence of COVID-19 is comprehensively considered and analyzed according to grades and credits. The proposed model can be used in teaching scenarios with a mixture of online and offline teaching, and its feasibility has been experimentally demonstrated with practical significance and application value.

In this paper, we first summarize and review the recent general methods for predicting and warning students' academic levels in courses. A neural network model for early warning of students' performance, which integrates their academic level, daily activities, course characteristics, and the changing circumstances of the teaching environment, is proposed to achieve the prediction of students' performance levels in universities to evaluate grades and to flag and warn students with poorly predicted grades. Finally, the model performance is evaluated experimentally and further analyzed on this result to discuss the error sources and possible directions for further optimization as well as the implications for practice.

II. RELATED WORKS

Multiple factors influence student achievement. While general thinking takes more into account the student's learning ability, specifically, there are many independent environmental distractions. Guo and Liu [7] analyzed in detail the main factors affecting grades based on real statistics at the undergraduate level in universities. The study showed that in addition to differences in students' abilities and teachers' teaching styles, the failure rate is correlated with the type of course. The authors also proposed an early warning model for grade monitoring based on the Goal, Management, Service, and Feedback (GMSF) modules, which form a closed-loop transaction processing with pre-, in-, and post-intervention, and can be applied to practical scenarios. Costa and Fleith [8] examined the scientific findings regarding the prediction of academic performance by cognitive and socio emotional variables from a psychological perspective through a systematic literature review of empirical articles and summarily argued for a significant impact of socio emotional (self-regulation, satisfaction, self-efficacy, and academic engagement) on student performance. Onu et al. [9] also predicted student performance in online courses from the perspective of student interest.

Currently, there are many programs and real-world examples of using data analytics tools, machine learning, and other techniques to assess and predict student performance, and the application of data mining techniques and machine learning approaches is widely used [10]. Mingyu et al. [4] first used a K-prototype-based construction of student portraits and then used Catboost-SHAP for grade prediction to achieve interpretable academic warning visualizations. Researchers such as Guo and Liu [7], Ma [11], implemented grade prediction in various practical scenarios from the perspective of multiple linear regression, binary logit regression, support vector machines, and neural networks, respectively. Yac [10] attempted to evaluate the performance of commonly

used machine learning algorithms for grade prediction applications, but the authors used only three dimensions of parameters.

Some other researchers have tried to adapt existing machine learning models or introduce new variables arising from student learning to track learning. Shi et al. [12] and others proposed a method for predicting objective and subjective scores based on cognitive diagnosis and neural networks, trying to solve the problem that score prediction can only be accurate to the overall subject score, but not the score of each subtopic, Knowledge states were used as features and student scores were used as training labels to predict the scores of objective and subjective questions separately. Xu et al. [13], on the other hand, considered correlation analysis between Internet behavior data and scores and found a strong correlation, which is in line with general laws and perceptions, while proposing an end-to-end two-layer self-attentive network that extracts the characteristics of students' Internet behavior over some time on the one hand and considers behavior over a longer time on the other. However, data extraction and analysis in this way rely on deep packet inspection (DPI) techniques and require interdisciplinary cooperation. Su et al. [14] improved the traditional data processing algorithm by proposing a cosine similarity-based course similarity calculation method and an improved hybrid multi-weight algorithm. Huang et al. [15] proposed an adaptive differential evolutionary neural network model and compared the fitness curves of genetic algorithms and differential evolutionary algorithms, but the study used a dataset with a capacity of only about 200, which is prone to over fitting, and the generalization ability needs further evaluation.

Long short-term memory neural networks (LSTM) are also often applied to achievement prediction because of their adeptness in analyzing inputs correlated with time series. This approach was used in the literature [16], [17], [18] proved to be more accurate than traditional models for academic ability prediction. Li and Gao [3] classified different dimensions, established a hierarchical process model, and used K-Means and genetic algorithms for data and weights were preprocessed, and then the results were introduced into a BP neural network in an attempt to predict students' English final grades, however, there are still problems such as insufficient data set capacity.

The global pandemic of COVID-19 has accelerated the popularity of online learning platforms (E-Leaning) [5], [19], while traditional prediction models for offline learning do not apply to the predictive assessment of online learning effectiveness, the literature [20], [21], [22] explored performance prediction approaches for online learning but were only able to perform performance prediction for learning on specific online platforms, and the authors did not discuss methods to integrate with traditional offline teaching performance prediction. At the same time, the dramatic shift in teaching methods has raised concerns among some researchers about the quality of teaching and learning [23], [24]. Since the level of knowledge and competence acquired through online learning is only 1/3 of that of traditional offline teaching [25]. Therefore, more researchers have also started to focus on the learning quality detection and performance warning mechanisms of online learning platforms. However, with many early warning models for offline teaching performance available in the past, there are few theoretical or practical engineering discussions on how online learning platforms can influence students' performance over a longer time and combine with offline teaching prediction models.

The cleaning and reliability of heterogeneous data has been an important factor affecting the stability of the model, limited by the reality of universities, where various types of systems work independently for a long time without relying on data middleware or datalakes. Redundant, repetitive data pre-processing is a consideration for almost every researcher, and a large amount of data (especially tags) needs to be pre-labeled manually. As the variety and total amount of behavioral data increases, identifying high-quality manual features becomes increasingly challenging. Jin and Zhu [26] first, they pointed out that the current data governance of smart campus big data platforms still suffers from fragmentation and redundancy, and designed a grade prediction classifier using KNN algorithm. In addition, Li and Wang [18] proposed a function that can automatically extract features from students' multi-source heterogeneous behavioral data to predict academic performance, which is based on the use of long- and short-term memory networks to capture the intrinsic time-series features of each behavior type.

More of the above scholars have focused their research mainly on predicting students' academic performance and thus providing early warnings to students who are expected to perform poorly, and the evaluation has also focused on validating existing data. However, Lu et al. [27] argue that neither a prediction nor early warning is the ultimate goal of the model. They argue that educators should be more committed to continuously improving students' abilities. To address this, they attempted to validate the contribution of early warning students to student achievement by providing continuous attention and early intervention.

Overall, by summarizing and categorizing the existing studies, predictions for student achievement and level are usually based on the following dimensional categories: students themselves (gender, enrolment performance, region of origin, type of origin), family environment (parents' occupation, income, level of education), historical achievement, and in some studies, even ethnicity and nationality are taken into consideration [28], however, this has the potential risk of discrimination. In addition, some common machine learning algorithms are effective, including linear regression, K-means, random forests, decision trees, support vector machines, and Bayesian decision-making, in addition to a small number of researchers who have also experimented with deep learning and neural networks, with some useful results.

III. DATA COLLECTION

Teaching management systems have been widely deployed in various higher education institutions in China with mature applications. Based on this, a unified smart campus platform has been derived through the integration of various information technology applications, such as monitoring and control management systems, financial systems, and Office Automation (OA) systems for IT infrastructure, which means that a large amount of data will be available for reference and use. Unfortunately, the initial raw data and model structure were designed for these above applications, not for this research. Therefore, in order to achieve our goal, it is necessary to pre-process the raw data, which converts the collected data into data that can be used in this study, and in fact, this work consumes more time. In this section, the source of the raw data and its structure will be explained first, followed by the structure of the data needed for the project. Finally, the process of cleaning and converting the data will be explained. Fig.1 shows the general table structure of a heterogeneous system database and the general principles of connection handling.

A. DATA SOURCES

The original data sources for the dataset used in this study were the university's academic registration management system, undergraduate academic affairs management system, library management system, and Access control management (AC). We collected records from the class of 2017 - 2021, spanning September 2017 to January 2022. Considering that the initial construction stages of the different systems did not use a unified database, all data were linked with the personal information within the academic registration management system. This was used to eliminate redundant fields.

B. DATA PREPARATION

First, the project team researched and analyzed the data in the database of each sub-module of the smart campus that might be used for analysis, classified the data by source, and designed SQL queries to correlate data items.

The data set input consists of three main parts: student performance statistics (R), student activities at school (A), External environmental information (E), and course statistics (S). Each table in the database does not use a particular attribute of an entity as the primary key but uses incremental record numbering, and the algebraic expression of the relationship between the three is (1):

$$\pi_D(\sigma_F(R \bowtie A \bowtie S \bowtie E)) \tag{1}$$

In relational algebra, σ represents selection, and the subscript is another relational algebraic expression to filter the entire row vector from the subsequent sets that meet the subscript condition. π represents projection, and the subscript is part of the filtered field data, thus eliminating redundant fields from the original data set.

In (1), the R is the historical grade distribution of each student. Specifically, it includes the student's gender, historical maximum score, minimum score, mean score, grade variance, and historical academic warning distribution. A is the student's daily activities during school, mainly including library visits and night-time Internet access, and the frequency of wireless Internet access near the apartment. S is the historical grade data for each course offered by the school, including the course's historical maximum score, minimum score, mean score, grade variance, and historical academic warning distribution. E is the external environment characteristics, which mainly cover external conditions that are not subjective factors and the will of the students. All four used the academic number and course as connecting conditions, with the final selection of all dimensions being D.

To ensure the validity of the data and to improve the usability of the dataset. need to filter extreme data. Based on this, the data is further cleaned by introducing the conditional expression in (1). The current discussion, F is mainly done as follows:

- Match field values using canonical regular expressions to filter out incorrect data, including incorrect student numbers and data that does not meet integrity constraints.
- Screening only grades for normal types of exams, excluding special grades such as make-up exams and retakes.
- Select only the valid grades after the normal marking and synthesis of grade types, provided that condition (2) is satisfied.
- From the target analysis,0 scores are generally for students who did not take the exam, cheated, or made mistakes in filling out personal information in the test papers and other personal special circumstances, so in such scenarios, it is no longer meaningful to conduct predictive analysis of their scores, and therefore, all 0 scores data are excluded.
- To balance the problem of inconsistent time scales of the cumulative statistical fields for different enrollment years, the project exported the data separately according to different enrollment years, followed by finding the average of their school semester as the normalized value, and finally merged the data sets.

According to the preliminary research, although a few related studies have considered the influence of the characteristics of the course itself on the distribution of student performance, the focus remains on the distribution of the same course across different classes in the same academic year and grade level. However, unlike mandatory public courses, some specialized courses have a small number of students taking them in the same academic year, which cannot generate a significant amount of data. This affects the prediction results and accuracy of these courses. Also, the teaching and grading habits of the same instructor for the same course are generally more consistent. Teacher psychological factors can also interfere with subjective ratings.

Therefore, unlike previous studies, for S in (1), this study no longer considers the grade distribution of the same class

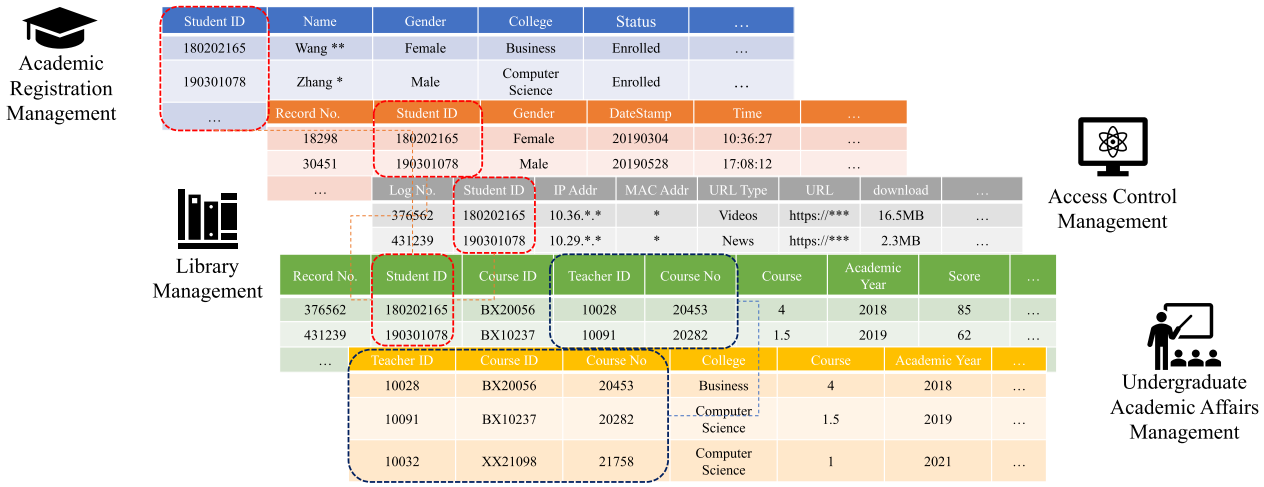


FIGURE 1. Raw heterogeneous data from multiple systems with significant redundancy, requiring multiple objects to be related using SQL queries, The student ID and course ID are the associated primary and foreign keys to focus on (The data in the figure are indicative only and do not represent some of the real data).

individually but uses a broader time scale, i.e., the same course, the same instructor, and the same credit course are considered uniformly, and the grade distribution of a course is analyzed comprehensively over a longer time, and Fig.2 shows the basis for grouping the data sets.

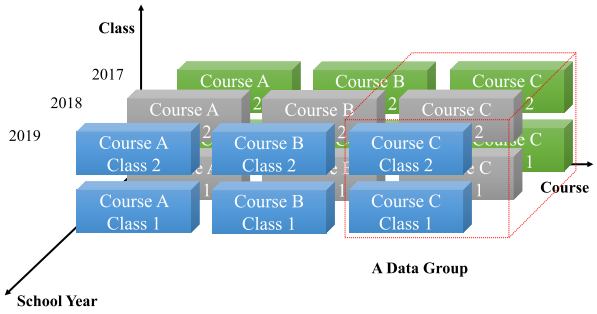


FIGURE 2. The basis of data grouping statistics.

Scores and rankings are important quantitative indicators of learning outcomes, and in this study, students' overall assessment scores for a particular course are placed in a set based on the above classification for the overall ranking, and the PERCENT_RANK method, which is widely used by most researchers, is used to map the grade rankings into the [0,1] interval so that different numbers of students have the same ranking scale for course rankings.

The library arrival information is reflected in the direction of the library gate entrance in the form of student ID identification records, so it is necessary to count each individual record separately. In the statistics of Internet behavior, according to the literature [13], late arrival and prolonged Internet browsing of non-study-related information has a significant negative correlation with students' academic performance. The Internet behavior management system does an excellent job in this regard. Thanks to deep packet inspection (DPI) and traffic mirroring techniques, network traffic can be collected and classified without affecting the performance of the campus network. Considering the importance of convenience and personal privacy protection, it is not necessary to collect specific applications and URLs in the scenario of this study; we usually only care about coarse-grained types of applications. In this study, entertainment, movies, and games accessed from the Internet between 23:00 and 6:00 from Monday to Friday were filtered from logs for statistics. The criterion for the statistics is the sum of the number of logs for the three mentioned above.

It should be noted that the predicted results are in semester cycles, and the counts counted need to be cut according to the time range, and thereafter, counted within each subset. To prevent the occurrence of excessively large polarized data, for the same statistical period, the average value of that count term is first calculated, i.e.(2):

$$\bar{s} = \frac{\sum_{i \in A} s_i}{N_A} \tag{2}$$

Then, the ratio of the statistical value to the mean is then used as the value among the data set, i.e.(3):

$$s_i' = \frac{s_i}{\bar{s}} \tag{3}$$

C. DATA TAGGING FOR E-EARNING

The wide spread of COVID-19 since the beginning of 2020 has posed a significant challenge to education efforts worldwide that were not anticipated by previous studies on the subject. In China and many parts of the world, students' failure to return to school on time will inevitably cause some of the statistical values that previous models relied on to not properly reflect the true situation, in addition to having an impact on their own academic performance. Based on this, this study considers the introduction of a COVID-19 flag in

the external environment, the value range of which is [0,1]. It is defined as the ratio of online distance learning to the number of class hours in the semester. We tried to introduce this flag bit to correct the model bias caused by the influence of COVID-19 and verified its feasibility in the experiment.

Fig.3 shows the relevant statistical proportions. For the COVID-19 Flag values, all values are taken to one decimal place on a statistical basis, except for the second semester of 2019-2020, which is taken to be 0.75 (the semester most affected by COVID-19), and thereafter the COVID-19 Flag is classified as 0.6, 0.4, and 0.3 for subsequent semesters. In addition to this, for data prior to 2019, where a small number of online courses exist, the COVID-19 Flag is specified as 0. These courses are not affected by the anomaly but are traditional online platform courses.

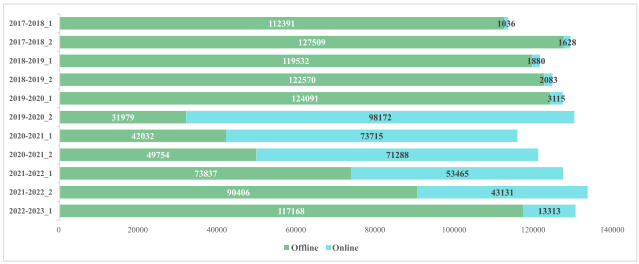


FIGURE 3. Relationship between the number of online and offline class hours between 2017 and 2023.

D. DATA SECURITY

This study was supported and assisted by the university's Network and Information Center, the Center for Modern Educational Research, the School of Computer and Information Security, and the School of Marxism, and as a result, access to relevant and important data was granted.

Within the raw data, there is some specific locational information such as student number, name, and other associated data (library access records, Internet behavior, historical grades, and any additional personal information). This information will be more sensitive compared to other instructional data. To avoid accessing and locating the original data as much as possible, while ensuring the reliability of the research results, the relevant data are strictly controlled and processed. Specifically:

- The association of data is necessary (which will ensure that each piece of data is logical). Still, the fields used for the association are unnecessary and more sensitive, and thus, the output data will not contain any information that can be directly located to an individual. The data filtered by (1) no longer includes the above, therefore each piece of data is anonymous.
- The data is pre-processed in a trusted, offline environment, and the SQL statements and scripts used are pre-secured by an audit system to ensure that the logic and implementation meet privacy requirements.
- The pre-processed data is also prohibited from being transferred through campus networks and other storage media. Model training is also done within the offline platform.

E. DATA DISTRIBUTION

Although some records may be missed after the above processing, the records retained are more reliable for further use. For example, for unrecorded data such as library visits, 0 can be used to fill in directly. Based on the above points, a total of more than 3.8 million records of various types were collected, and after cleaning, merging, linking, and integrating, more than 1 million valid records were collected as the dataset to be used. Unlike most existing studies that use only a small amount of data for modeling, the data package used in this study includes the grades of all undergraduate courses for all four years at the university.

Fig.4(a) shows the distribution of raw scores, while Fig.4(b) reflects the distribution of warning levels further determined by raw scores.

In addition, the Rising Sun plot in Fig.4(c) shows the distribution of data volume across grades and semesters. It should be noted that there are significantly more courses in the lower grades of the undergraduate program than in the upper grades, which is due to the fact that the upper grades contain more practical aspects and inquiry processes and a longer learning cycle. Note that the ".5" is used to distinguish between two semesters in a full academic year.

IV. MODEL DESIGN AND METHODS

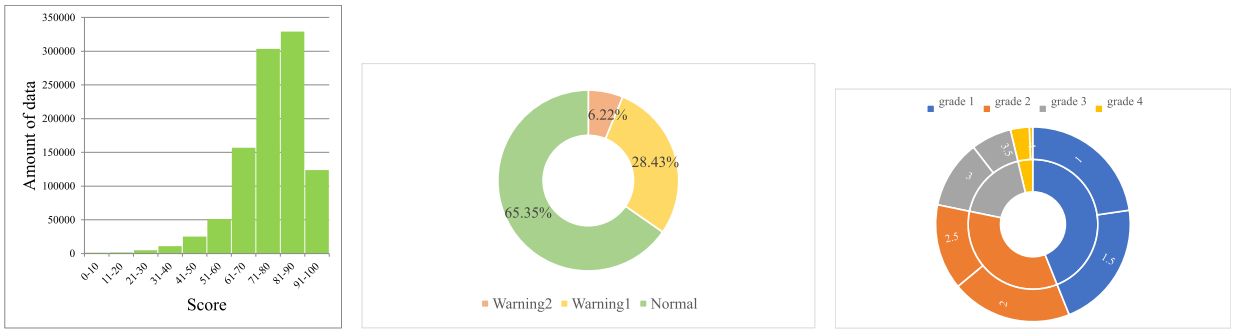
Fig.5 shows the overall architecture of the model and the working operation levels. The raw data of the heterogeneous system belongs to the data layer, which is collected and cleaned as data to be used. The SMOTE process enhances the sensitivity of the model to a few classifications. At the same time, the model is suitably generalized with a cost-sensitive learning algorithm to adapt it to multi-classification scenarios.

The model accepts the processed inputs and produces the corresponding outputs. In the research evaluation phase, the shared prediction data is used, while in the production environment, the updated data is used.

This section will first introduce the inputs and outputs of the model, then explain the SMOTE process and the multi-classification cost-sensitive learning hybrid optimization method, and finally explain the structure and parameters of the model.

A. INPUT AND OUTPUT

Table.1 provides a categorical summary of the input dimensions used. Personal information mainly includes gender as well as second-level colleges and departments. For grade data, not only the historical grade distribution of students is considered, but also the historical grade distribution characteristics of the courses within a certain range are counted. Daily activities were analyzed for library access data as well as internet usage, and the external environment covered course credit distribution and COVID-19 Flag.



(a) distribution of raw scores (b) distribution of warning levels further determined by raw scores (c) distribution of data volume across grades and semesters

FIGURE 4. Pre-processed data distribution.

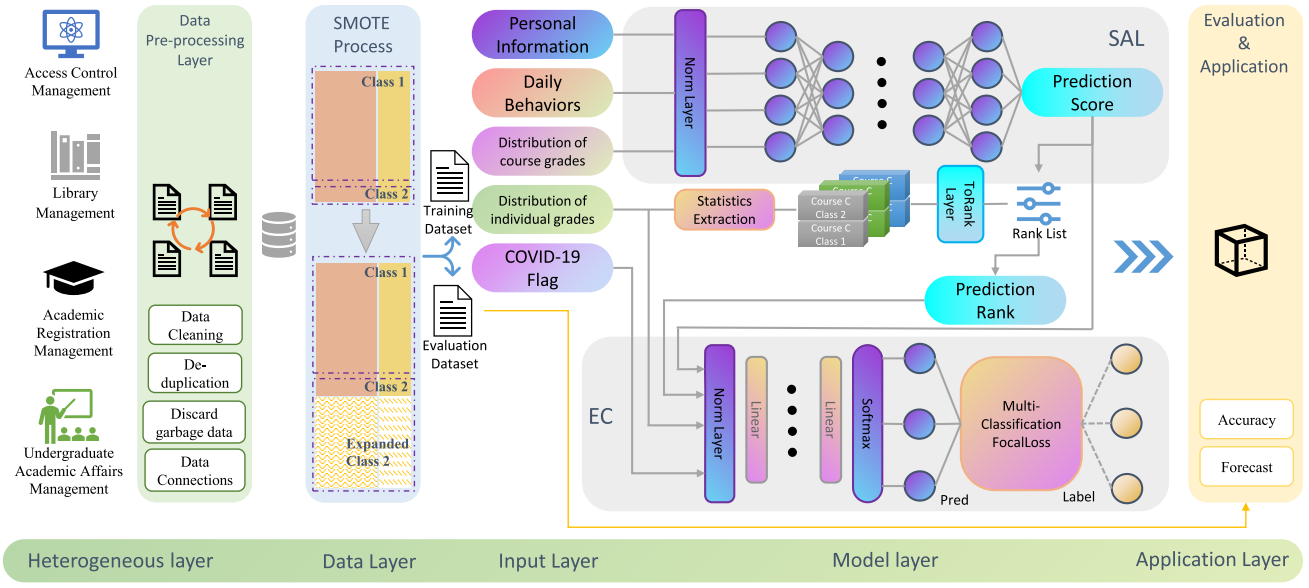


FIGURE 5. Model overall operating structure.

It is important to note that the data indicated by Course History score distribution is a data set shown in Fig.2 as a data set those outputs individual statistical values. Credits are listed separately in External Environment to consider that the same course has different student requirements for different faculties and students. The same course with different credits also belongs to different statistical groups.

For the output, the score prediction is simply the predicted value of the subject score for the given interval. The actual situation of the school is also considered: In Chinese schools, test scores are usually on a percentage scale, with 60 being a pass. This study considers the practical implications and inevitable errors of the early warning mechanism and uses 65 as the early warning threshold.

For score prediction, it is not necessary to obtain accurate scores in most cases; instead, educators are more interested in the distribution of scores across the curriculum and the grades of student performance. Therefore, we defined "Grade Warning Levels"from the actual range of scores and labeled them with a one-hot vector. Table.2 illustrates the relationship between the overall score and the warning level.

B. SMOTE PROCESS

In the data categories of this study, the label distribution is unbalanced. In fact, similar problems exist in similar studies, which can cause the prediction results to produce spuriously high accuracy while masking the true situation [29]. SMOTE (Synthetic Minority Over-sampling Technique) is a technique used to balance the category imbalance in the data set. The basic idea is to increase the number of minority samples by synthesizing some new minority samples to balance the dataset. Fig.6 shows the SMOTE process in action.

Specifically, SMOTE first randomly selects a sample x from a small number of categories C and calculates its Euclidean distance d from N similar proximity points. Then, new samples can be generated by constructing

TABLE 1. Input Dimension and ClassificationType.

Dimensions Type	Indicators
Personal Information	Gender Colleges or Departments
Student History score Distribution	Highest score Minimum Score Average score Variance Warning 1 (Attention) Frequency Warning 2 (Warning) Frequency
Course History score Distribution	Highest score Average score Variance Warning 1 (Attention) Frequency Warning 2 (Warning) Frequency Ranking
Daily Activities	Library Access Internet Access
External Environment	Course Credits Grade Level Affected by COVID-19(E-Leaning) Flag

TABLE 2. Output Classification.

Warning Level	Intervals	One-Hot vector
Normal	(65,100]	[1,0,0]
Warning 1 (Attention)	[50,65]	[0,1,0]
Warning 2 (Warning)	(0,50)	[0,0,1]

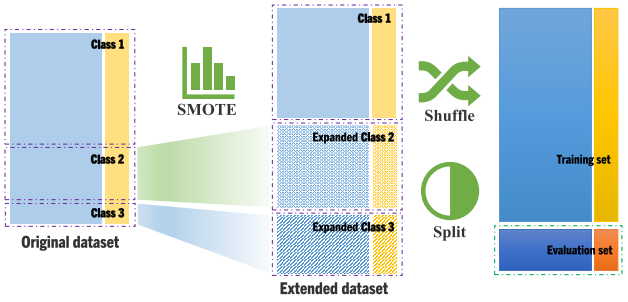


FIGURE 6. SMOTE process principle.

according to (4).

$$x_{new} = x + \alpha \cdot d \tag{4}$$

For the original data, the data distribution was first analyzed to identify the types of labels with significantly less data. Next, a SMOTE operation is performed on the small category data, thus extending its number. After that, it is merged with the original data for the shuffle operation, and the dataset is divided at the same time, thus ensuring that the split dataset is equally balanced.

It should be noted that although down-sampling can also compensate for the imbalance of small classification samples to some extent, it can also lead to the loss of a large amount of feature information. Meanwhile, SMOTE can generate new data to increase the amount of data while retaining the original data, which can improve the performance of the classifier, and the added data perturbation helps to reduce the risk of over fitting the model, thus improving the generalization ability of the classifier.

C. COST-SENSITIVE MULTICLASS CLASSIFICATION

In general classification tasks, the cross-entropy loss is widely used and considered robust. However, as shown in Fig.4(b), the data labeled as "Attention" and "Warning" represent a small percentage of the entire dataset, which leads the classifier to always focus on the characteristics of the majority of samples The SMOTE process balances the number of samples for each classification in the data layer. In contrast, the introduction of cost-sensitive learning FocalLoss is a practical application [30], which improves the cross-entropy loss of the traditional SoftMax layer by not only considering the weights of different samples, but also focusing on those samples that are more difficult to classify and is calculated as in (5).

$$FocalLoss = -\alpha_t (1 - p_t)^{\gamma} \ln p_t \tag{5}$$

where p_t is a variable that measures the ease of classification, and its calculation in the classification problem is as in (6).

$$p_t = \begin{cases} P(\widehat{y}) & \widehat{y} = y \\ 1 - P(\widehat{y}) & otherwise \end{cases} \tag{6}$$

Analysis of (6) shows that pt tends to be close to the value of 0 whenever the predicted category is different from the labeled category, while closer to 0 indicates that the group of data is more difficult to be distinguished by the model. It is generally believed that the model predicts this sample easily and credibly when $p_t \geq 0.6$ [30]. γ and α denote the attenuation coefficients on the original cross-entropy and the weights of each category in the labels.

In addition, during the training process of deep learning models, the problem of gradient disappearance or gradient explosion easily occurs, while Focal Loss reduces the influence of these samples by reducing the weights of easily classified samples, thus making the model more stable.

FocalLoss can also be applied to multi-classification, but it needs to be modified for the characteristics of multi classification tasks. In this study, FocalLoss is appropriately extended to adapt to multi-categorization based on its design idea. The p_t value of each category is calculated separately, and then the weighted average loss value is calculated according to the weight of each category set by α, which is calculated as in (7). The specific logic can be found in the pseudocode shown in Algorithm.1.

$$FocalLoss = -\frac{1}{N} \sum_{t} \alpha_t (1 - p_t)^{\gamma} \ln p_t \tag{7}$$

D. MODEL DESIGN

To fully consider the influence of students' learning ability and external factors on students' performance and improve the effectiveness of early warning, a SAL&EC based on Student Academic Level (SAL) and External Environmental

Algorithm 1 Extended Cost-Sensitive Multiclass Classification

Input: The prediction result vector:
$\boldsymbol{P} = (p_0, p_1, \ldots, p_{n-1})$;
Label One-Hot vector:
$\boldsymbol{L} = (l_0, l_1, \ldots, l_{n-1})$;
P_t values for each classification;
Weight vector for classification:
$\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \ldots, \alpha_{n-1})$

Output: *FocalLoss*

LossSUM $= 0$
for $i = 0; i < n; i++$ **do**
  if $L[i] = 1$ **then**
    $P_t[i] = P[i]$
  else
    $P_t[i] = 1 - P[i]$
  end if
  $LossSUM += \alpha[i] * [(1 - P_t[i])^{\gamma}] * \log P_t[i]$
end for
$FocalLoss = -LossSUM/n$
return *FocalLoss*

Characteristics (EC) is designed. The SAL network first generates a prediction of the student's overall assessment score based on the student's historical academic level and course characteristics, which is an accurate prediction. Thereafter, the EC network combines the results of the SAL network while introducing various external environmental features that do not depend on the student's own will to enhance the model's understanding of the external objective environment. Finally, the prediction results are transformed into probability distributions using a SoftMax layer enhanced by multi-classification sensitive cost learning to provide early warning decisions. The ''model layer''in Fig.5 shows the overall structure of the SAL&EC network.

The SAL network focuses on the prediction of course scores and consists of one normalized layer and four hidden layers, using students' academic ability statistics and personal characteristics, and daily behaviors as inputs, with the output being the predicted scores.

The scores predicted by the SAL network have some errors. In fact, we demonstrate this in the experiments in Section V. In order to further increase the credibility of the model, this study designed an environmental perception network based on External Characteristics (EC), which accepts external information such as academic year, grade level, course distribution data over multiple terms, and course credits, while using the output of the SAL network as the basis for analysis.

In addition, the project team researched the grading habits of some teachers and the school regulations related to ranking as one of the correlates of student performance. However, the situation was different from the training concentration. There is a strict causal logical relationship between grades and rankings, and in the production environment, the ranking of that grade is not available for input until a definite score is obtained. Nevertheless, once a predicted grade is obtained, it can still be entered into the Ranking Layer to obtain the position of that grade in the historical grade distribution.

Between SAL network and EC network. Statistics extraction extracts the total assessment scores of all eligible courses from Course score Distribution as the historical scores of the category and ranks them in descending order. Thereafter ToRankLayer will analytically compute the historical ranking of the predicted scores and map them to a Distribution in [0,1]. Algorithm 2 and Algorithm 3 describe the computational process of StatisticsExtraction and ToRankLayer, respectively.

Algorithm 2 Statistics Extraction

Input: Original dataset $D^{L\times W}$,
Filter criteria F, Dimension number to be sorted t.

Output: S^l_{DESC}

extract the l rows from $D^{L\times W}$ that match condition F to form the objective matrix, i. e: $D^{l\times W}_{obj} = \pi_W \sigma_F \left(D^{L\times W}\right)$
Defining the characteristic unit matrix:
$E_W = (\alpha_0, \alpha_1, \alpha_2, \ldots, \alpha_{W-1})$
Calculation $S^{l\times 1} \Leftarrow D^{l\times W}_{obj} \alpha^t$
Sort $S^{l\times 1}$ in descending order and transform to get descending sequence S^l_{DESC}

Algorithm 3 ToRankLayer

Input: Predicted value $\widehat{y}$,
consisting of labeled values S^l_{DESC}
Fulfillment:
$S^l_{DESC} = \{s_0, s_1, \ldots, s_{l-1} \mid s_0 \geq s_1 \geq \ldots \geq s_{l-1}\}$;

Output: *Rank*

if $l > 1$ **then**
  for s_i in S^l_{DESC} **do**
    if $\widehat{y} < s_i$ **then**
      $i++$
    else
      Break
    end if
  end for
  $Rank = \frac{i-1}{l-1}$
else if $l = 1$ **then**
  if $\widehat{y} < s_0$ **then**
    $Rank = 1$
  else
    $Rank = 0$
  end if
else if $l = 0$ **then**
  $Rank = 0$
end if
return *Rank*

The environment-aware network with external features has one normalization layer and four hidden layers, which

introduces the external environment while considering course grades and student grade distributions and transforms the prediction results into a one-hot vector of warning levels through predictions into a one-hot vector of warning levels by a SoftMax layer optimized by multi classification cost-sensitive learning, thus realizing the early warning of academic performance.

For the choice of other parameters, the learning rate was set to 10^{-5} and the BatchSize to 128. ReLU() is a common activation function, but in multiple hidden layers it may cause the subsequent neurons to fail to continue forward with valid information due to premature zeroing of neurons [31], so RReLU() or Leaky ReLU() was used to avoid this. Where, α for Leaky ReLU is taken as 0.01. The SAL network loss function uses SmoothL1Loss(). Compared with MAE (Mean Absolute Error) and MSE (Mean Square Error), it both improves the problem of unsmooth at the zero point of MAE and is not too sensitive and aggressive as MSE when the error value is too large. The calculation principle is as in (8),0.5 is a widely used empirical parameter.

$$SmoothL1Loss(x) = \begin{cases} 0.5x^2 & |x| < 1 \\ |x| - 0.5 & |x| > 1 \end{cases} \tag{8}$$

The optimizer uses Adam, which has become very popular in the last two years. It not only considers the momentum update but also incorporates the dynamic adjustment of the learning rate of each dimension. It is an adaptive learning rate optimization algorithm that combines the advantages of two other optimization algorithms: AdaGrad and RMSProp [32]. The basic idea behind the Adam optimizer is to use adaptive learning rates for each parameter. It maintains an exponentially decaying average of past gradients, called the first moment, and an exponentially decaying average of past squared gradients, called the second moment. These two moments are then used to calculate the update to the parameters in each iteration.

Its weight update formula is shown in (9), and the Adam optimizer updates the parameters based on the gradient of each parameter and the exponential moving average of the historical gradient. In each iteration, Adam calculates estimates of the first-order moments (i.e., mean) and second-order moments (i.e., variance) of the current gradient and uses these values to calculate the current step size. In this way, Adam can adapt to different scales for different parameters, thus accelerating convergence. Where $\widehat{m}_t$ and $\widehat{m}_t$ in (9) denote the first-order exponential smoothing value of the historical gradient and the first-order exponential smoothing value of the squared historical gradient, respectively, and are calculated as in (10) and (11) to obtain the gradient values with momentum and the learning rate weight parameter for each weight parameter. In the experiments of this study, β_1 is taken as 0.9 and β_2 as 0.96.

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\widehat{v}_t + \epsilon}} \cdot \widehat{m}_t \tag{9}$$

$$\widehat{m}_t = \frac{m_t}{1 - \beta_1^t} \tag{10}$$

$$\widehat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{11}$$

V. RESULTS AND DISCUSSION

A. ASSESSMENT BASIS AND INDICATORS

In machine learning, performance evaluation for classification problems often uses Accuracy, Precision, and Recall. Accuracy is a metric that measures the performance of a classification model by calculating the proportion of correctly classified instances among all instances in the dataset. It is typically expressed as a percentage, and the higher the accuracy score, the better the model's performance. In the binary classification problem, the formula for calculating accuracy in (12):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

where TP (true positives) is the number of instances that are correctly classified as positive, TN (true negatives) is the number of instances that are correctly classified as negative, FP (false positives) is the number of instances that are incorrectly classified as positive, and FN (false negatives) is the number of instances that are incorrectly classified as negative.

Precision is the fraction of correctly predicted positive instances among all predicted positive instances. It is defined as(13):

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

Precision measures how well the model can identify the positive instances, and it is useful when the cost of false positives is high. For example, In the present study, a false positive (Good grades in a course but categorized as requiring attention) can be very costly, In the use of real-life scenarios, educational administrators may therefore mistakenly focus on students who do not require special attention, resulting in a waste of educational resources.

Recall, on the other hand, measures the proportion of positive instances that are correctly identified by the model. It is also known as sensitivity or true positive rate. Recall is calculated as follows (14):

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

Recall is particularly useful in situations where the cost of false negatives (i.e., instances that are incorrectly classified as negative) is high, especially in our study, more attention needs to be paid to achievement anomalies, a high recall rate is desired, even if it comes at the expense of lower accuracy.

The above three parameters can be generalized to multi-classification problems. In multi-classification, the accuracy rate is the ratio of the total number of correctly predicted samples to the total number of predictions. Furthermore, for each category, accuracy and recall are independent of each other. Obviously, in multi-classification, each classification

has an independent precision and recall rate. Specifically, it is expressed:

- For Precision: the proportion of all samples predicted as category C that are correctly classified.
- For Recall: the proportion of all samples predicted correctly by the classifier for category C.

Precision and Recall are contradictory metrics; in general, Recall tends to be low when Precision is high, while Recall tends to be high when Precision is low. When the classification confidence is high, Precision is high; when the classification confidence is low, Recall is high. To be able to consider these two metrics together, F-measure is proposed (a weighted summed average of Precision and Recall), i.e.(15):

$$F_1 = 2 \times \frac{P \cdot R}{P + R} \tag{15}$$

The core idea of F_1 is to improve Precision and Recall as much as possible while also wanting the difference between them to be as small as possible. F_1 applies to Binary classification problems. For multi-classification problems, the F_1 of dichotomous is generalized to have two metrics, $Micro_{F1}$ and $Macro_{F1}$:

- $Micro_{F1}$: $Micro_{F1}$ takes into account the total number of true positives, false positives, and false negatives across all classes, and calculates the F_1 score based on those counts. This means that each instance contributes equally to the final score, regardless of its class label.
- $Macro_{F1}$: $Macro_{F1}$, on the other hand, calculates the F_1 scores for each class separately, and then takes the average of those scores. This means that each class contributes equally to the final score, regardless of its size.

In summary, The F_1 value is a common metric to evaluate classifiers with values between [0,1], which tries to combine the precision and recall of each classification to give a more objective performance parameter. And F_1 is divided into $Macro_{F1}$ and $Micro_{F1}$. $Micro_{F1}$ considers the number of each class, however, $Macro_{F1}$ does not consider the number of data, so it will look at each class equally. In this study, we used $Macro_{F1}$, which is calculated as follows (16):

$$Macro_{F1} = \frac{1}{n} \sum_{i=1}^{n} F_i \tag{16}$$

B. ANALYSIS OF TEST RESULTS

In this study, the College was most severely impacted by COVID-19 in the spring 2020 semester when, according to our survey, More than 70% of the teaching hours are conducted through online courses, and this study used that semester as the baseline for all data generated by the COVID-19 flag with a flag value of 0.75. Thereafter, we investigated online instruction and student return dates for subsequent semesters to determine their range. It should be noted that for records before 2020, the COVID-19 flag was set to 0.

From the confusion matrix of the validation set results, the accuracy and recall of each warning level was calculated as shown in Table.3, where the overall prediction accuracy was 94.09%, which achieved the desired goal.

To verify the validity of the model, several sets of ablation experiments were performed in this study. The models were re-evaluated separately using only one of these networks. First, the EC network was excluded from the SAL&EC network, and only the Student Academic Level (SAL) network with the SoftMax layer was used, and the same training data was used and tested on the validation set, and the confusion matrix of the experimental results is shown in Table.3. Then, only the EC network was used, and the experimental results are shown in Table.3. It can be clearly seen that considering only a single student subjective factor with external environmental factors, its early warning accuracy is inferior to both SAL & EC-BP networks.

To further assess the ability of the EC network to correct the original prediction values, this study also considered the direct mapping of the SAL network prediction values to the warning levels and compared the results with those of the SAL & EC-BP networks. Specifically, the prediction scores of the SAL network, based on the exclusion of the EC network, were used, and directly converted to the early warning levels according to Table.1. The accuracy of each warning level is shown in Fig.7

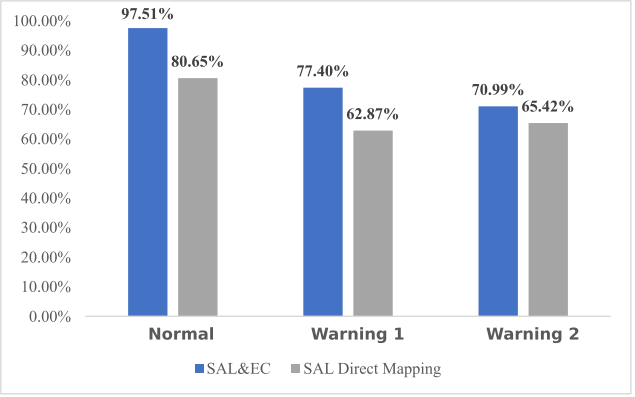


FIGURE 7. Direct mapping of SAL network prediction results versus structure using EC network.

The accuracy of early warning can be improved to some extent after the adoption of the EC network to sense the external environment. It also shows that various factors affect students' performance; besides students' ability and personal habits, the external environment and learning stage also have some influence on students' performance. In general, most people prefer to believe that academic performance is a highly subjective activity, but this study also explains to some extent the influence of objective factors such as teachers and course difficulty on students' final grades.

This study also assessed the prediction accuracy of the same class of students over a long period. Specifically, data from the prediction set were taken separately for the class

TABLE 3. Comparison of accuracy and recall of prediction results for different networks.

Network	Labels	Accuracy	Recall	MacroF1	overall	Macro-F1
SAL&EC	Normal	97.51%	97.66%	0.9759	94.09%	0.8221
	Attention	77.40%	75.31%	0.7634		
	Warning	70.99%	74.47%	0.7269		
Only SAL	Normal	84.96%	83.02%	0.8398	81.76%	0.5638
	Attention	55.46%	30.67%	0.3950		
	Warning	59.29%	37.12%	0.4566		
Only EC	Normal	63.89%	59.07%	0.6139	61.03%	0.5223
	Attention	43.63%	37.52%	0.4034		
	Warning	58.22%	52.04%	0.5496		
Fully connected	Normal	89.12%	84.54%	0.8677	84.98%	0.7620
	Attention	73.28%	67.94%	0.7051		
	Warning	73.02%	69.71%	0.7133		

of Grade 1 and the class of Grade 2, divided by semester, and tested separately for prediction accuracy in different academic years. The results are shown in Fig.8 and Fig.9. Thereafter, we also re-labeled some of the predicted data to ignore the effect of the COVID-19(E-Learning) Flag field value. In Fig.8 and Fig.9, the background intervals colored with darker colors indicate the extent to which semester instruction is affected by COVID-19, generally considering mainly the second semester of the 2019-2020 academic year. The results show that the EC network that considers the impact of COVID-19 can somewhat improve the accuracy. Also, the EC network has the most significant improvement in prediction accuracy in the semester most affected by the COVID-19 epidemic, due to many changes in teaching methods and the impact of long home hours, online courses, and library suspensions.

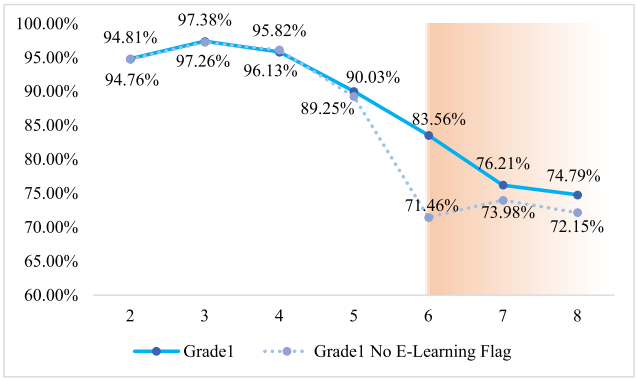


FIGURE 8. Grade1 students, the trend of prediction accuracy over the same period.

We also relabeled some of the predicted data in an attempt to ignore the effect of the COVID-19 Flag field value. Specifically, in Fig.8 and Fig.9, the background intervals colored in dark colors indicate the extent to which semester instruction is affected by COVID-19.In general, we primarily consider the second semester of the 2019-2020 academic year and set the COVID-19 Flag to 0.75 using this as a benchmark, and thereafter the COVID-19 Flag is classified as 0.6, 0.4, and 0.3 for subsequent semesters, which are calculated according to The values are taken according to the calculation mentioned in

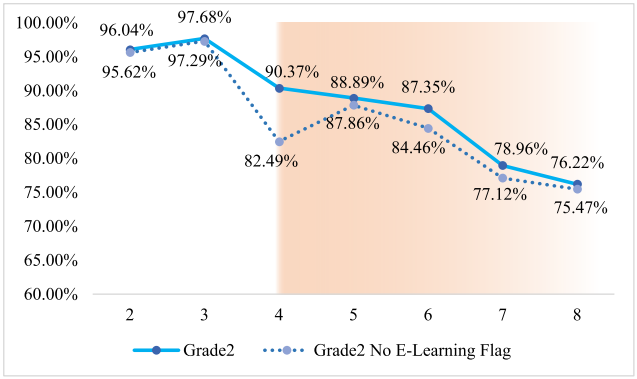


FIGURE 9. Grade2 students, the trend of prediction accuracy over the same period.

the III. In Fig.8 and Fig.9, the solid line reflects the accuracy of the prediction in this scenario, while the dashed line indicates the accuracy of the prediction after excluding the value of the COVID-19 flag, i.e., setting this field of the prediction to 0. The practical implication is that it is assumed that the teaching method in these semesters is not different from that before 2020 and is offline, which does not correspond to the actual situation.

The results show that the EC network that takes into account the impact of COVID-19 can improve accuracy to some extent, especially in the spring 2020 semester, which is the most affected by the COVID-19 epidemic, where the EC network has the most significant improvement in prediction accuracy due to a large number of changes in the way courses are taught, the impact of students being home for long periods, online courses, and the suspension of library access. In a deeper analysis, COVID-19 Flag directly or indirectly influenced the distribution of many data. Through experimental verification, with the influence of COVID-19 Flag excluded, $Macro_{F1}$ is 0.7216, and after introducing this in the EC network, $Macro_{F1}$ is raised to 0.8221, further illustrating the importance of the EC network. Also, this set of experiments demonstrates another perspective on the COVID-19 outbreak epidemic and the profound impact of the epidemic prevention policy on teaching and learning in mainland China universities.

In some similar studies, most researchers have used common machine-learning algorithms for performance prediction. Therefore, this study also designed experiments to compare SVM, decision tree, and KNN for prediction separately, and the results are shown in Table.4 comparing SAL&EC in Table.3, it can be seen that the accuracy and recall of SAL&EC for Normal classification were improved by about 7%, but due to the EC network and SortingLayer, ToRankLayer The accuracy and recall of the Attention and Warning categories were significantly improved by the introduction of the EC network and the SortingLayer and ToRankLayer, which are the key to intervene in the whole early warning system for academics in the production environment. Therefore, the performance optimization at the Warning level is more significant compared to the model's small improvement in Normal classification.

In a similar study, the XGBoost-FM model was also used to predict whether a student would pass the exam [33], to analyze whether there was a risk of failing the course in which the student took the exam. The ultimate goal of the model is a dichotomous set, and the original dataset labels can be used in this model after modification. Meanwhile, we also compare a parallel CNN (Convolutional Neural Networks) based fusion network prediction model (Hybrid 2D CNN) [34]. The model characteristics are first populated with one-dimensional data and then constructed in two dimensions, which leads to channel splitting and dimensionality enhancement. Multiple chains are finally fused through multiple parallel max-pooling and convolutional layers. Table.5 compares the performance of XGoost-FM, Hybrid 2D CNN, and the model in this paper in two and three classifications.

The $Macro_{F1}$ of the XGBoost-FM model for triclassification is 0.6231, corresponding to an AUC of 0.913 for dual classification, while the $Macro_{F1}$ of Hybrid 2D CNN is 0.6205 and the AUC is 0.896. Under the same conditions, the $Macro_{F1}$ of SAL&EC-BP in three classifications is 0.8221 and the AUC is 0.965.

The ROC curves of the above models and algorithms are compared in Fig.10.The experimental results show that both XGBoost-FM and SAL&EC have good prediction ability and practical value, but SAL&EC improves the model prediction level significantly in the application of multi-level warning. In addition, in Fig.10, the AUC values of KNN, decision tree, and SVM are 0.825, 0.887, and 0.873.

The long-time span prediction performance of several other models and algorithms for two consecutive classes is shown in Fig.11. In the same trend as in Fig.8 and Fig.9, the performance of all models and algorithms tested tended to decrease when analyzed for upper undergraduate classes, which is analyzed at the end of this section.

The data associated with this study spans a long period of time. New data has been generated weekly since the project began. To further evaluate the prediction level of the model in the current scenario. We extracted some of the new data

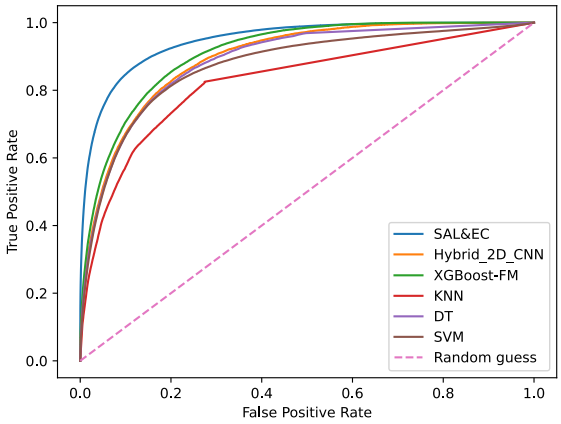


FIGURE 10. ROC curve comparison of the above various models and algorithms.

since the fall of 2022 and made predictions using the three methods described above, with the confusion matrix shown in Fig.12. Of these valid scores: 59.37%, 30.08%, and 10.55% for the three levels of Normal, Attention, and Warning, which is generally consistent with the historical data distribution shown in Fig.4(b).

Analyzing the experiments and errors, there was a tendency for the accuracy of predictions to decrease progressively over four years for the same cohort, and we analyzed the possible causes of this phenomenon.

- The number of courses in the first and second years of undergraduate studies is high, and they are mostly public courses and professional foundation courses, and the teaching format is also more traditional. For such scenarios, the model fitting ability is strong. From the third year onward, with the introduction of professional courses and experimental courses, the teaching format and grading criteria are more flexible, the original way of assessing students' characteristics needs to be further expanded, and the distribution of grades in historical courses needs to be more fully considered in terms of students' interest in the major, career planning and participation in the professional field.
- Starting from the third year of undergraduate studies, some Chinese students begin to prepare for the master's degree entrance examinations and library visits rise, and their grades in in-class courses are limited due to their different goals, while other students participate in various practical studies, internships, etc., and these grading criteria are quite different from those of in-class courses. Therefore, further separate discussions are needed but also rely on more detailed questionnaires and data.

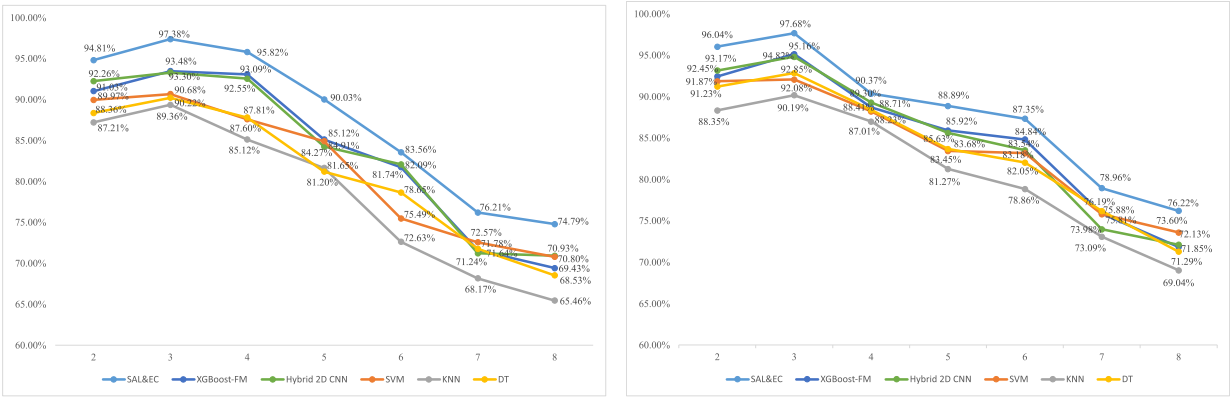
Indeed, the introduction of an external environment is a challenging task that requires constant attention. 2021 is considered a critical year for the beginning of the "post-COVID-19 era" [35] (for mainland China, the second half of 2022). Instructional reform and school development may

TABLE 4. Comparison of accuracy and recall of prediction results of different Machine Learning Algorithms.

Network	Labels	Accuracy	Recall	F1	Macro-F1
SVM	Normal	88.13%	98.40%	0.9298	0.6595
	Attention	56.11%	22.56%	0.3218	
	Warning	70.99%	74.47%	0.7269	
Decision Trees	Normal	90.64%	89.51%	0.9007	0.5383
	Attention	34.46%	36.71%	0.3555	
	Warning	34.69%	37.11%	0.3586	
KNN	Normal	89.68%	94.88%	0.9221	0.5637
	Attention	44.53%	31.39%	0.3682	
	Warning	50.99%	33.03%	0.4009	

TABLE 5. Comparing the performance of the model in this paper and the XGBoost-FM model under two and three classifications.

Model	Category Number	Labels	Accuracy	Recall	F1	Macro-F1
XGBoost-FM	2	Normal	91.27%	96.97%	0.9403	-
		Warning	73.12%	57.44%	0.6434	
	3	Normal	93.38%	71.84%	0.8121	0.6231
		Attention	52.59%	51.07%	0.5182	
		Warning	49.65%	58.93%	0.5389	
Hybrid 2D CNN	2	Normal	90.74%	93.46%	0.9208	-
		Warning	69.97%	62.26%	0.6589	
	3	Normal	90.15%	72.36%	0.8028	0.6205
		Attention	56.08%	50.65%	0.5323	
		Warning	53.73%	54.24%	0.5399	
SAL&EC	2	Normal	94.92%	97.03%	0.9596	-
		Warning	80.55%	73.38%	0.7680	
	3	Normal	97.51%	97.66%	0.9759	0.8221
		Attention	77.40%	75.31%	0.7634	
		Warning	70.99%	74.47%	0.7269	

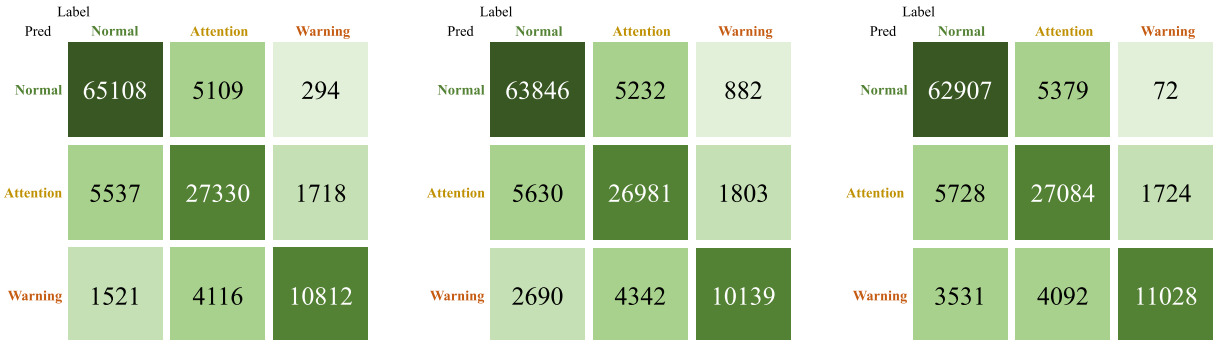


(a) Grade1 students, the trend of prediction accuracy over the same period.

(b) Grade2 students, the trend of prediction accuracy over the same period.

FIGURE 11. Long time span prediction performance of several other models and algorithms in two consecutive terms.

turn this change into a new normal. For example, according to the findings in Fig.3, in the most recent semester, even though it appears that instruction has returned to normal (meaning 2023), the number of hours and the percentage of instruction using E-Learning still reached about 10%, significantly higher than the previous less than 2%. Some teachers believe that although the widespread implementation of E-Learning is a program in a particular context, it has also allowed them to discover the characteristics and advantages of this teaching method and to be able to combine it with various new technologies to produce better results. Therefore, they partially continued this approach and applied it in their teaching practices. However, for this study, we need to think about extending the E-Leanning Flag or considering other external environmental influences (e.g., bringing in data from some online learning platforms) in order to adapt to these ''new normals'' and thus achieve better performance of the model.



(a) SAL&EC. (b) XGBoost-FM. (c) Hybrid 2D CNN.

FIGURE 12. Comparison of prediction performance of the above three models on more recent data since fall 2022.

It is true that the introduction of more external environmental features can make the model more adaptable to real-world situations, but it also increases the complexity of the model at the same time, and a large amount of directional data also raises moral and ethical controversies about data security and data collection. However, we believe that deep learning is still a solution that can continue to be investigated and used because it is good at mining the intrinsic connections from within a limited data dimension while avoiding the direct collection of more information.

In addition, the introduction of SortingLayer and ToRankLayer is considered to have a greater impact on model training. It should be noted that the adoption of static rank instead of the full SortingLayer and ToRankLayer is a sacrifice of partial accuracy for the improvement of training efficiency.

VI. CONCLUSION

In this paper, we propose a SAL&EC neural network that considers students' academic level and external environmental characteristics. The student's daily school behavior is utilized, while the student's historical academic level and the course's historical grade distribution are considered to mine the possibility of a specific student passing the course. In order to apply external environmental factors to the performance warning, an environment-aware network (EC) was designed with special consideration of the impact of E-learning on college teaching and learning to improve prediction accuracy. The SAL&EC neural network employs a data-level SMOTE approach while incorporating multi-classification cost-sensitive weights to handle imbalanced data and improve the correct rate of hard-to-classify samples. The experimental results show that the SAL&EC neural network model has good accuracy in student grade early warning and has a robust application value.

In addition, we will pay close attention to changes in the external environment and seek to adapt the model to the new teaching system, while relying less on individual-oriented data and focusing more on factors that indirectly affect individuals.

In the future, we will consider classifying course types and using different dimensions for flexible evaluation for practical courses, online courses, etc., to improve the accuracy of grade prediction for upper-level undergraduate courses. Currently, the SAL network relies on historical student grade information, which makes it impossible to effectively assess students' grades in their first semester of enrolment. Of course, there have been some attempts at this problem, such as the use of high school-level grades, richer information about students' backgrounds, and entrance exam results.

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